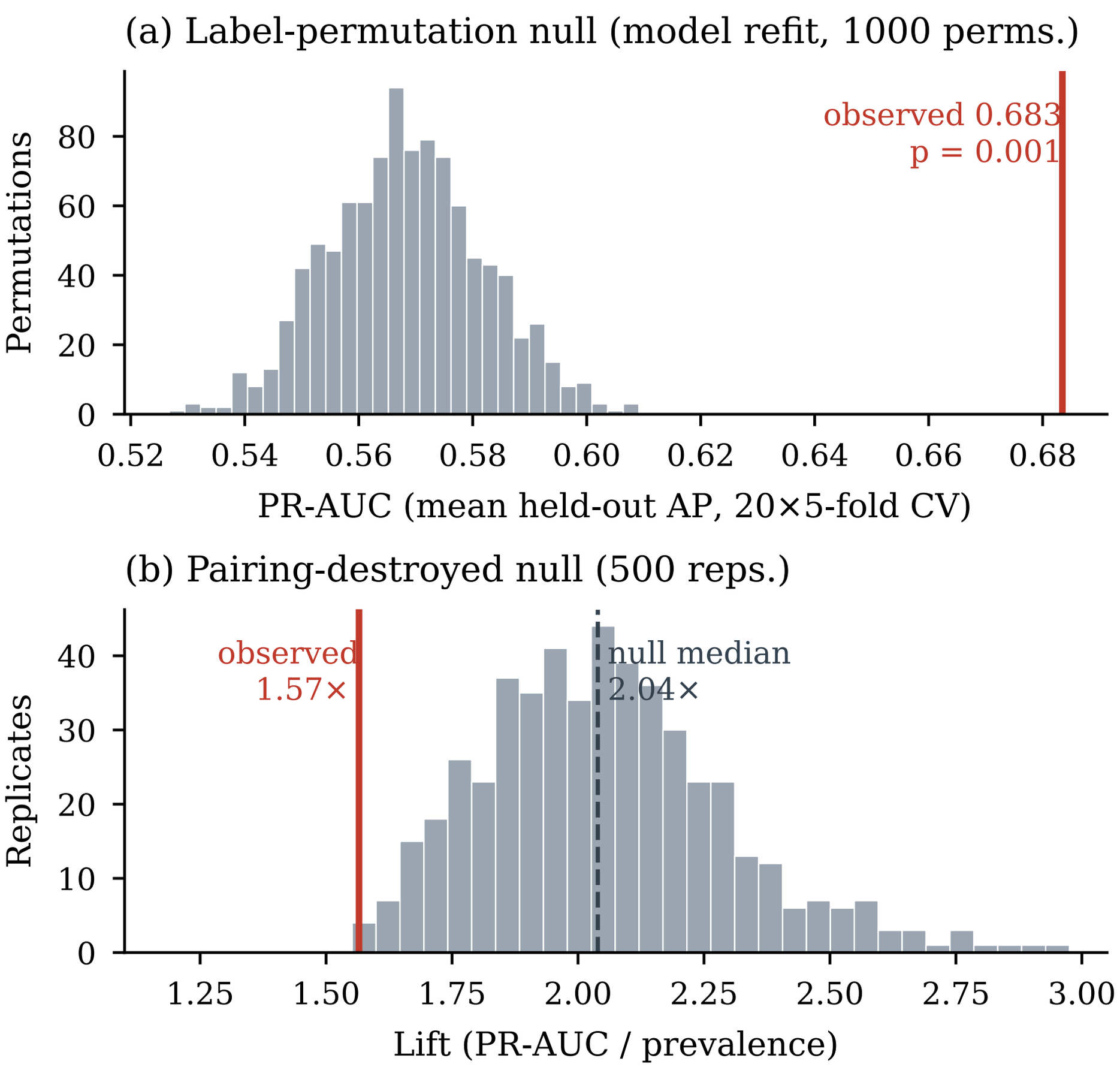
(a) Label-permutation null (model refit, 1000 perms.)
observed 0.683
p = 0.001
Permutations
0
20
40
60
80
0.52
0.54
0.56
0.58
0.60
0.62
0.64
0.66
0.68
PR-AUC (mean held-out AP, 20×5-fold CV)
(b) Pairing-destroyed null (500 reps.)
observed
1.57×
null median
2.04×
Replicates
0
10
20
30
40
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00
Lift (PR-AUC / prevalence)